

In Problems 27–30, a partial differential equation (PDE) is given along with the form of a solution having separated variables. Show that such a solution must satisfy the indicated set of ordinary differential equations.

27. $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$

with $u(r, \theta) = R(r)\Theta(\theta)$ yields

$$r^2 R''(r) + rR'(r) - \lambda R(r) = 0,$$

$$\Theta''(\theta) + \lambda \Theta(\theta) = 0,$$

where λ is a constant.

28. $\frac{\partial^2 u}{\partial t^2} + \frac{\partial u}{\partial t} + u = \alpha^2 \frac{\partial^2 u}{\partial x^2}$

with $u(x, t) = X(x)T(t)$ yields

$$X''(x) + \lambda X(x) = 0,$$

$$T''(t) + T'(t) + (1 + \lambda \alpha^2)T(t) = 0,$$

where λ is a constant.

29. $\frac{\partial u}{\partial t} = \beta \left\{ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right\}$

with $u(x, y, t) = X(x)Y(y)T(t)$ yields

$$T'(t) + \beta \lambda T(t) = 0,$$

$$X''(x) + \mu X(x) = 0,$$

$$Y''(y) + (\lambda - \mu)Y(y) = 0,$$

where λ, μ are constants.

30. $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} = 0$

with $u(r, \theta, z) = R(r)\Theta(\theta)Z(z)$ yields

$$\Theta''(\theta) + \mu \Theta(\theta) = 0,$$

$$Z''(z) + \lambda Z(z) = 0,$$

$$r^2 R''(r) + rR'(r) - (r^2 \lambda + \mu)R(r) = 0,$$

where μ, λ are constants.

31. For the PDE in Problem 27, assume that the following boundary conditions are imposed:

$$u(r, 0) = u(r, \pi) = 0,$$

$$u(r, \theta) \text{ remains bounded as } r \rightarrow 0^+.$$

Show that a nontrivial solution of the form $u(r, \theta) = R(r)\Theta(\theta)$ must satisfy the boundary conditions

$$\Theta(0) = \Theta(\pi) = 0,$$

$$R(r) \text{ remains bounded as } r \rightarrow 0^+.$$

32. For the PDE in Problem 29, assume that the following boundary conditions are imposed:

$$u(0, y, t) = u(a, y, t) = 0; \quad 0 \leq y \leq b, \quad t \geq 0,$$

$$\frac{\partial u}{\partial y}(x, 0, t) = \frac{\partial u}{\partial y}(x, b, t) = 0; \quad 0 \leq x \leq a, \quad t \geq 0.$$

Show that a nontrivial solution of the form $u(x, y, t) = X(x)Y(y)T(t)$ must satisfy the boundary conditions

$$X(0) = X(a) = 0,$$

$$Y'(0) = Y'(b) = 0.$$

33. When the temperature in a wire reaches a steady state, that is, when u depends only on x , then $u(x)$ satisfies Laplace's equation $\partial^2 u / \partial x^2 = 0$.

(a) Find the steady-state solution when the ends of the wire are kept at a constant temperature of 50°C , that is, when $u(0) = u(L) = 50$.

(b) Find the steady-state solution when one end of the wire is kept at 10°C , while the other is kept at 40°C , that is, when $u(0) = 10$ and $u(L) = 40$.

10.3 Fourier Series

While analyzing heat flow and vibrating strings in the previous section, we encountered the problem of expressing a function in a trigonometric series [compare equations (12) and (25) in Section 10.2]. In the next two sections, we discuss the theory of Fourier series, which deals with trigonometric series expansions. First, however, we review some function properties that are particularly relevant to this study: piecewise continuity, periodicity, and even and odd symmetry.

In Section 7.2 we defined a **piecewise continuous** function on $[a, b]$ as a function f that is continuous at every point in $[a, b]$, except possibly for a finite number of points at which f has a *jump discontinuity*. Such functions are necessarily integrable over any finite interval on which they are piecewise continuous.

Recall also that a function is **periodic of period T** if $f(x + T) = f(x)$ for all x in the domain of f . The smallest positive value of T is called the **fundamental period**. The trigonometric functions $\sin x$ and $\cos x$ are examples of periodic functions with fundamental period 2π and $\tan x$ is periodic with fundamental period π . A constant function is a periodic function with arbitrary period T .

Two symmetry properties of functions will be useful in the study of Fourier series. A function f that satisfies $f(-x) = f(x)$ for all x in the domain of f has a graph that is symmetric with respect to the y -axis [see Figure 10.4(a)]. We say that such a function is **even**. A function f that satisfies $f(-x) = -f(x)$ for all x in the domain of f has a graph that is symmetric with respect to the origin [see Figure 10.4(b)]. It is said to be an **odd** function. The functions $1, x^2, x^4, \dots$ are examples of even functions, while the functions $x, x^3, x^5, \dots$ are odd. The trigonometric functions $\sin x$ and $\tan x$ are odd functions and $\cos x$ is an even function.

Example 1 Determine whether the given function is even, odd, or neither.

(a) $f(x) = \sqrt{1 + x^2}$ (b) $g(x) = x^{1/3} - \sin x$ (c) $h(x) = e^x$

Solution

- (a) Since $f(-x) = \sqrt{1 + (-x)^2} = \sqrt{1 + x^2} = f(x)$, then $f(x)$ is an even function.
 (b) Because $g(-x) = (-x)^{1/3} - \sin(-x) = -x^{1/3} + \sin x = -(x^{1/3} - \sin x) = -g(x)$, it follows that $g(x)$ is an odd function.
 (c) Here $h(-x) = e^{-x}$. Since $e^{-x} = e^x$ only when $x = 0$ and e^{-x} is never equal to $-e^x$, then $h(x)$ is neither an even nor an odd function. ♦

Knowing that a function is even or odd can be useful in evaluating definite integrals. The following result, illustrated in Figure 10.4, is a straightforward consequence of the definition of the definite integral.

Properties of Symmetric Functions

Theorem 1. If f is an even piecewise continuous function on $[-a, a]$, then

(1)
$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

If f is an odd piecewise continuous function on $[-a, a]$, then

(2)
$$\int_{-a}^a f(x) dx = 0.$$

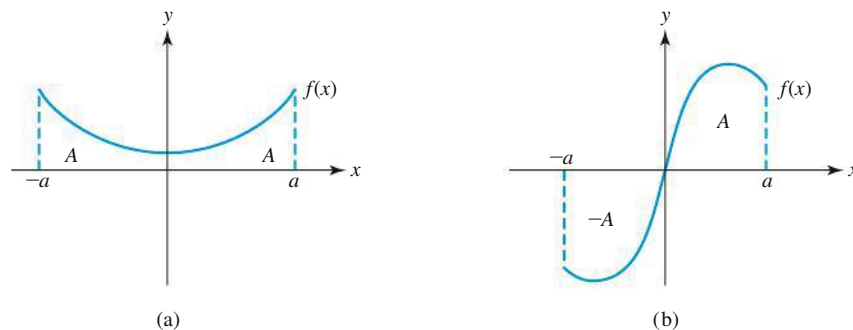


Figure 10.4 (a) Even function $\int_{-a}^a f = A + A = 2 \int_0^a f$ (b) Odd function $\int_{-a}^a f = A - A = 0$

The next example deals with certain integrals that are crucial in Fourier series.

Example 2 Evaluate the following integrals when m and n are nonnegative integers:

$$\begin{aligned} \text{(a)} \quad & \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx, & \text{(b)} \quad & \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx, \\ \text{(c)} \quad & \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx. \end{aligned}$$

Solution The even and odd functions occurring in the integrands are sketched in Figure 10.5.

The given integrals are easily evaluated by invoking the trigonometric formula for products of sines and cosines:

$$\begin{aligned} \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} &= \frac{1}{2} \sin \frac{(m-n)\pi x}{L} + \frac{1}{2} \sin \frac{(m+n)\pi x}{L}, \\ \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} &= \frac{1}{2} \cos \frac{(m-n)\pi x}{L} - \frac{1}{2} \cos \frac{(m+n)\pi x}{L}, \\ \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} &= \frac{1}{2} \cos \frac{(m-n)\pi x}{L} + \frac{1}{2} \cos \frac{(m+n)\pi x}{L}. \end{aligned}$$

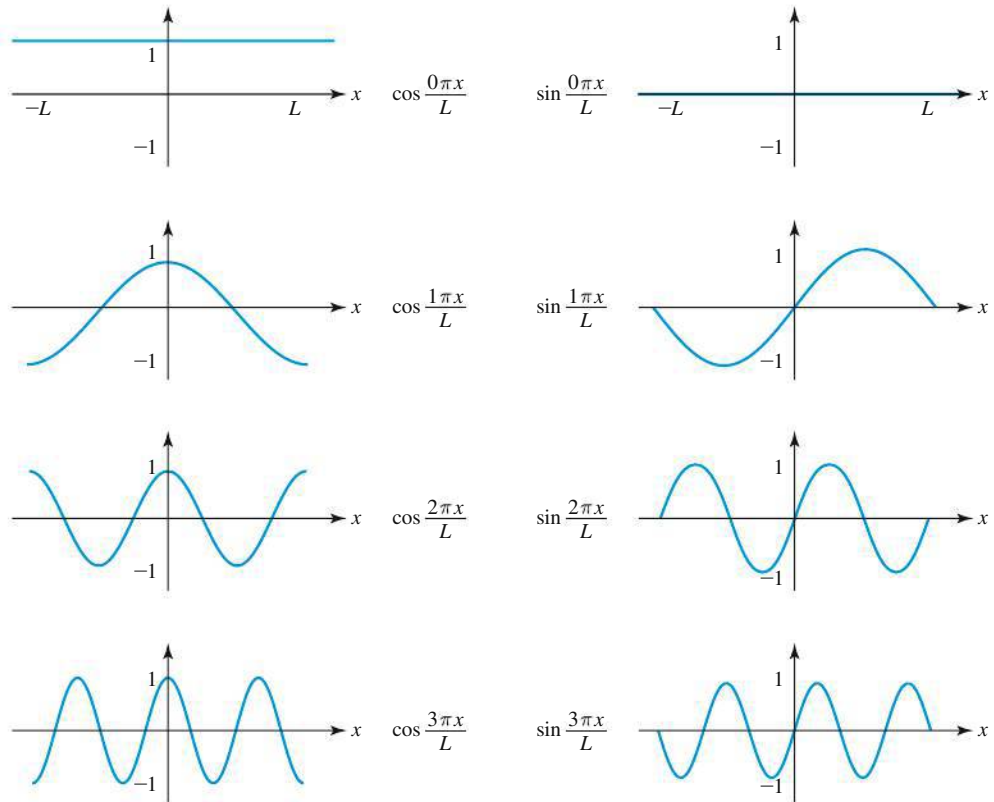


Figure 10.5 The sinusoids

Thus if $m \neq n$, each of the integrals calls for the (signed) area under an oscillating sinusoid over a whole number of periods, and is therefore zero. In fact, the only way to avoid these “oscillators” is to take $m = n$; whence $\cos[(m-n)\pi x/L] = \cos 0 = 1$ and subtends an area of $2L$, while $\sin[(m-n)\pi x/L] = \sin 0 = 0$ subtending zero area (again). But note that if m and n are *both* zero, then $\cos[(m+n)\pi x/L] = \cos 0$ also subtends area $2L$. Restoring the factors of $1/2$, we summarize with

$$(3) \quad \int_{-L}^L \sin \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = 0,$$

$$(4) \quad \int_{-L}^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx = \begin{cases} 0, & m \neq n, \\ L, & m = n, \end{cases}$$

$$(5) \quad \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \begin{cases} 0, & m \neq n, \\ L, & m = n \neq 0, \\ 2L, & m = n = 0. \end{cases} \quad \blacklozenge$$

Equations (3)–(5) express an **orthogonality condition**[†] satisfied by the set of trigonometric functions $\{1 = \cos 0x, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$, where $L = \pi$. We will say more about this later in this section.

It is easy to verify that if each of the functions $f_1, \dots, f_n$ is periodic of period T , then so is any linear combination

$$c_1 f_1(x) + \dots + c_n f_n(x).$$

For example, the sum $7 + 3 \cos \pi x - 8 \sin \pi x + 4 \cos 2\pi x - 6 \sin 2\pi x$ has period 2, since each term has period 2. Furthermore, if the infinite series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

consisting of $2L$ -periodic functions converges for all x , then the function to which it converges will be periodic of period $2L$.

Just as we can associate a Taylor series with a function that has derivatives of all orders at a fixed point, we can identify a particular trigonometric series with a piecewise continuous function. To illustrate this, let's assume that $f(x)$ has the series expansion[‡]

$$(6) \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\},$$

where the a_n 's and b_n 's are constants. (Necessarily, f has period $2L$.)

To determine the coefficients $a_0, a_1, b_1, a_2, b_2, \dots$, we proceed as follows. Let's integrate $f(x)$ from $-L$ to L , assuming that we can integrate term by term:

$$\int_{-L}^L f(x) dx = \int_{-L}^L \frac{a_0}{2} dx + \sum_{n=1}^{\infty} a_n \int_{-L}^L \cos \frac{n\pi x}{L} dx + \sum_{n=1}^{\infty} b_n \int_{-L}^L \sin \frac{n\pi x}{L} dx.$$

[†]This nomenclature is suggested by the fact that the formulas for the Riemann sums approximating the integrals (3)–(5) look like dot products (Section 9.1, page 496) of higher-dimensional vectors. In most calculus texts, it is shown that the dot product of orthogonal vectors in two dimensions is zero.

[‡]The choice of constant $a_0/2$ instead of just a_0 will be motivated shortly.

The (signed) area under the “oscillators” is zero. Hence,

$$\int_{-L}^L f(x) dx = \int_{-L}^L \frac{a_0}{2} dx = a_0 L,$$

and so

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx.$$

(Notice that $a_0/2$ is the average value of f over one period $2L$.) Next, to find the coefficient a_m when $m \geq 1$, we multiply (6) by $\cos(m\pi x/L)$ and integrate:

$$(7) \quad \int_{-L}^L f(x) \cos \frac{m\pi x}{L} dx = \frac{a_0}{2} \int_{-L}^L \cos \frac{m\pi x}{L} dx + \sum_{n=1}^{\infty} a_n \int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx \\ + \sum_{n=1}^{\infty} b_n \int_{-L}^L \sin \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx.$$

The orthogonality conditions (3)–(5) render the integrals on the right-hand side quite immediately. We have already observed that

$$\int_{-L}^L \cos \frac{m\pi x}{L} dx = 0, \quad m \geq 1,$$

and, by formulas (3) and (5), we have

$$\int_{-L}^L \sin \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = 0, \\ \int_{-L}^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L} dx = \begin{cases} 0, & n \neq m, \\ L, & n = m. \end{cases}$$

Hence, in (7) we see that only one term on the right-hand side survives the integration:

$$\int_{-L}^L f(x) \cos \frac{m\pi x}{L} dx = a_m L.$$

Thus, we have a formula for the coefficient a_m :

$$a_m = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{m\pi x}{L} dx.$$

Similarly, multiplying (6) by $\sin(m\pi x/L)$ and integrating yields

$$\int_{-L}^L f(x) \sin \frac{m\pi x}{L} dx = b_m L$$

so that the formula for b_m is

$$b_m = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{m\pi x}{L} dx.$$

Motivated by the above computations, we now make the following definition.

Fourier Series

Definition 1. Let f be a piecewise continuous function on the interval $[-L, L]$. The **Fourier series**[†] of f is the trigonometric series

$$(8) \quad f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\},$$

where the a_n 's and b_n 's are given by the formulas[‡]

$$(9) \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 0, 1, 2, \dots,$$

$$(10) \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx, \quad n = 1, 2, 3, \dots$$

Formulas (9) and (10) are called the **Euler-Fourier formulas**. We use the symbol $\sim$ in (8) to remind us that this series is associated with $f(x)$ but may not converge to $f(x)$. We will return to the question of convergence later in this section. Let's first consider a few examples of Fourier series.

Example 3 Compute the Fourier series for

$$f(x) = \begin{cases} 0, & -\pi < x < 0, \\ x, & 0 < x < \pi. \end{cases}$$

Solution Here $L = \pi$. Using formulas (9) and (10), we have

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x dx = \frac{x^2}{2\pi} \Big|_0^{\pi} = \frac{\pi}{2},$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_0^{\pi} x \cos nx dx \\ &= \frac{1}{\pi n^2} \int_0^{\pi n} u \cos u du = \frac{1}{\pi n^2} [\cos u + u \sin u] \Big|_0^{\pi n} \\ &= \frac{1}{\pi n^2} (\cos n\pi - 1) = \frac{1}{\pi n^2} [(-1)^n - 1], \quad n = 1, 2, 3, \dots, \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{\pi} x \sin nx dx \\ &= \frac{1}{\pi n^2} \int_0^{\pi n} u \sin u du = \frac{1}{\pi n^2} [\sin u - u \cos u] \Big|_0^{\pi n} \\ &= \frac{-\cos n\pi}{n} = \frac{(-1)^{n+1}}{n}, \quad n = 1, 2, 3, \dots \end{aligned}$$

[†]*Historical Footnote:* Joseph B. J. Fourier (1768–1830) developed his series for solving heat flow problems. Lagrange expressed doubts about the validity of the representation, but Dirichlet devised conditions that ensured its convergence. Note that the notational choice of $a_0/2$ in equation (6) made it unnecessary to insert an extra formula for a_0 in (9).

[‡]Notice that $f(x)$ need not be defined for every x in $[-L, L]$; we need only that the integrals in (9) and (10) exist.

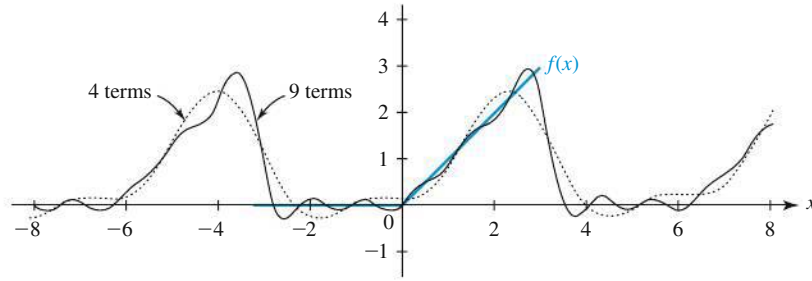


Figure 10.6 Partial sums of Fourier series in Example 3

Therefore,

$$\begin{aligned}
 (11) \quad f(x) &\sim \frac{\pi}{4} + \sum_{n=1}^{\infty} \left\{ \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx + \frac{(-1)^{n+1}}{n} \sin nx \right\} \\
 &= \frac{\pi}{4} - \frac{2}{\pi} \left\{ \cos x + \frac{1}{9} \cos 3x + \frac{1}{25} \cos 5x + \cdots \right\} \\
 &\quad + \left\{ \sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \cdots \right\}.
 \end{aligned}$$

Some partial sums of this series are displayed in Figure 10.6. ♦

Example 4 Compute the Fourier series for

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi. \end{cases}$$

Solution Again, $L = \pi$. Notice that f is an odd function. Since the product of an odd function and an even function is odd (see Problem 7), $f(x) \cos nx$ is also an odd function. Thus,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx = 0, \quad n = 0, 1, 2, \dots$$

Furthermore, $f(x) \sin nx$ is the product of two odd functions and therefore is even, so

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi} \sin nx \, dx \\
 &= \frac{2}{\pi} \left[\frac{-\cos nx}{n} \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{1}{n} - \frac{(-1)^n}{n} \right], \quad n = 1, 2, 3, \dots, \\
 &= \begin{cases} 0, & n \text{ even}, \\ \frac{4}{\pi n}, & n \text{ odd}. \end{cases}
 \end{aligned}$$

Thus,

$$(12) \quad f(x) \sim \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n} \sin nx = \frac{4}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \cdots \right].$$

Some partial sums of (12) are sketched in Figure 10.7 on page 578. ♦

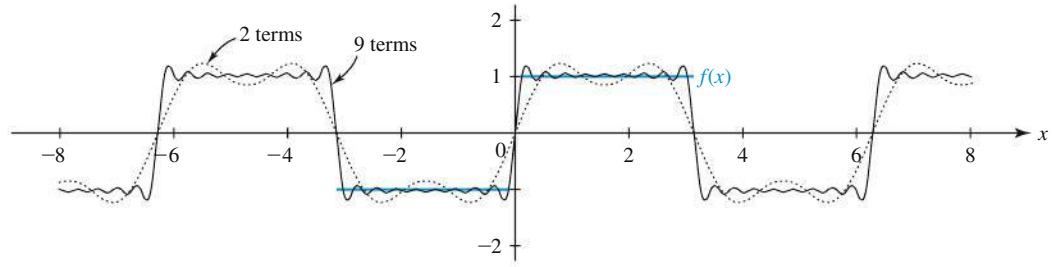


Figure 10.7 Partial sums of Fourier series in Example 4

In Example 4 the odd function f has a Fourier series consisting only of sine functions. It is easy to see that, in general, if f is any odd function, then its Fourier series consists only of sine terms.

Example 5 Compute the Fourier series for $f(x) = |x|$, $-1 < x < 1$.

Solution Here $L = 1$. Since f is an even function, $f(x) \sin(n\pi x)$ is an odd function. Therefore,

$$b_n = \int_{-1}^1 f(x) \sin(n\pi x) dx = 0, \quad n = 1, 2, 3, \dots$$

Since $f(x) \cos(n\pi x)$ is an even function, we have

$$a_0 = \int_{-1}^1 f(x) dx = 2 \int_0^1 x dx = x^2 \Big|_0^1 = 1,$$

$$\begin{aligned} a_n &= \int_{-1}^1 f(x) \cos(n\pi x) dx = 2 \int_0^1 x \cos(n\pi x) dx \\ &= \frac{2}{\pi^2 n^2} \int_0^{\pi n} u \cos u du = \frac{2}{\pi^2 n^2} [\cos u + u \sin u] \Big|_0^{\pi n} = \frac{2}{\pi^2 n^2} (\cos n\pi - 1) \\ &= \frac{2}{\pi^2 n^2} [(-1)^n - 1], \quad n = 1, 2, 3, \dots \end{aligned}$$

Therefore,

$$\begin{aligned} (13) \quad f(x) &\sim \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi^2 n^2} [(-1)^n - 1] \cos(n\pi x) \\ &= \frac{1}{2} - \frac{4}{\pi^2} \left\{ \cos(\pi x) + \frac{1}{9} \cos(3\pi x) + \frac{1}{25} \cos(5\pi x) + \dots \right\}. \end{aligned}$$

Partial sums for (13) are displayed in Figure 10.8. ♦

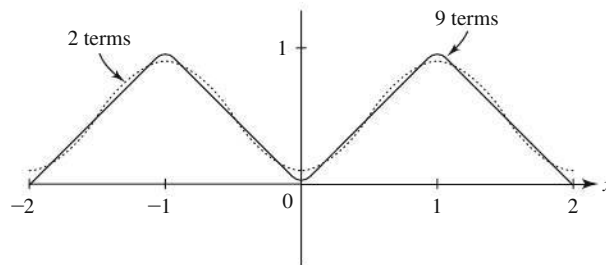


Figure 10.8 Partial sums of Fourier series in Example 5

Notice that the even function f of Example 5 has a Fourier series consisting only of cosine functions and the constant function $1 = \cos(0\pi x)$. In general, if f is an even function, then its Fourier series consists only of cosine functions [including $\cos(0\pi x)$].

Orthogonal Expansions

Fourier series are examples of orthogonal expansions.[†] A set of functions $\{f_n(x)\}_{n=1}^{\infty}$ is said to be an **orthogonal system** or just **orthogonal** with respect to the nonnegative weight function $w(x)$ on the interval $[a, b]$ if

$$(14) \quad \int_a^b f_m(x)f_n(x)w(x)dx = 0, \quad \text{whenever } m \neq n.$$

As we have seen, the set of trigonometric functions

$$(15) \quad \{1, \cos x, \sin x, \cos 2x, \sin 2x, \dots\}$$

is orthogonal on $[-\pi, \pi]$ with respect to the weight function $w(x) \equiv 1$. If we define the **norm** of f as

$$(16) \quad \|f\| := \left[\int_a^b f^2(x)w(x)dx \right]^{1/2},$$

then we say that a set of functions $\{f_n(x)\}_{n=1}^{\infty}$ (or $\{f_n(x)\}_{n=1}^N$) is an **orthonormal system with respect to** $w(x)$ if (14) holds and also $\|f_n\| = 1$ for each n . Equivalently, we say the set is an orthonormal system if

$$(17) \quad \int_a^b f_m(x)f_n(x)w(x)dx = \begin{cases} 0, & m \neq n, \\ 1, & m = n. \end{cases}$$

We can always obtain an orthonormal system from an orthogonal system just by dividing each function by its norm. In particular, since

$$\int_{-\pi}^{\pi} \cos^2 nx dx = \int_{-\pi}^{\pi} \sin^2 nx dx = \pi, \quad n = 1, 2, 3, \dots$$

and

$$\int_{-\pi}^{\pi} 1 dx = 2\pi,$$

then the orthogonal system (15) gives rise on $[-\pi, \pi]$ to the orthonormal system

$$\{(2\pi)^{-1/2}, \pi^{-1/2} \cos x, \pi^{-1/2} \sin x, \pi^{-1/2} \cos 2x, \pi^{-1/2} \sin 2x, \dots\}.$$

If $\{f_n(x)\}_{n=1}^{\infty}$ is an orthogonal system with respect to $w(x)$ on $[a, b]$, we might ask if we can expand a function $f(x)$ in terms of these functions; that is, can we express f in the form

$$(18) \quad f(x) = c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) + \dots$$

for a suitable choice of constants $c_1, c_2, \dots$? Such an expansion is called an **orthogonal expansion**, or a **generalized Fourier series**.

[†]Orthogonality is also discussed in Section 8.8 on page 474.

To determine the constants in (18), we can proceed as we did in deriving Euler's formulas for the coefficients of a Fourier series, this time using the orthogonality of the system. Presuming the representation (18) is valid, we multiply by $f_m(x)w(x)$ and integrate to obtain

$$\begin{aligned} (19) \quad \int_a^b f(x)f_m(x)w(x)dx &= c_1 \int_a^b f_1(x)f_m(x)w(x)dx + c_2 \int_a^b f_2(x)f_m(x)w(x)dx + \cdots \\ &= \sum_{n=1}^{\infty} c_n \int_a^b f_n(x)f_m(x)w(x)dx. \end{aligned}$$

(Here we have also assumed that we can integrate term by term.) Because the system is orthogonal with respect to $w(x)$, every integral on the right-hand side of (19) is zero except when $n = m$. Solving for c_m gives

$$(20) \quad c_m = \frac{\int_a^b f(x)f_m(x)w(x)dx}{\int_a^b f_m^2(x)w(x)dx} = \frac{\int_c^b f(x)f_m(x)w(x)dx}{\|f_m\|^2}, \quad n = 1, 2, 3, \dots$$

The derivation of the formula for c_m was only *formal*, since the question of the convergence of the expansion in (18) was not answered. If the series $\sum_{n=1}^{\infty} c_n f_n(x)$ converges *uniformly* to $f(x)$ on $[a, b]$, then each step can be justified, and indeed, the coefficients are given by formula (20). The notion of uniform convergence is discussed in the next subsection and in Section 13.2.[†]

Convergence of Fourier Series

Let's turn to the question of the convergence of a Fourier series. For Example 5 it is possible to use a comparison or limit comparison test to show that the series is absolutely dominated by a p -series of the form $\sum_{n=1}^{\infty} 1/n^2$, which converges. However, this is much harder to do in Example 4, since the terms go to zero like $1/n$. Matters can be even worse, since there exist Fourier series that *diverge*.[‡] We state two theorems that deal with the convergence of a Fourier series and two dealing with the properties of termwise differentiation and integration. For proofs of these results, see *Partial Differential Equations of Mathematical Physics*, 2nd ed., by Tyn Myint-U (Elsevier North Holland, Inc., New York, 1983), Chapter 5; *Advanced Calculus with Applications*, by N. J. DeLillo (Macmillan, New York, 1982), Chapter 9; or an advanced text on the theory of Fourier series.

Before proceeding, we need a notation for the left- and right-hand limits of a function. Let

$$f(x^+) := \lim_{h \rightarrow 0^+} f(x+h) \quad \text{and} \quad f(x^-) := \lim_{h \rightarrow 0^+} f(x-h).$$

We now present the fundamental pointwise convergence theorem for Fourier series. While reading it, keep in mind Figures 10.6-10.8.

[†]All references to Chapters 11-13 refer to the expanded text, *Fundamentals of Differential Equations and Boundary Value Problems*, 7th ed.

[‡]In fact, there are trigonometric series that converge but are not Fourier series; an example is $\sum_{n=1}^{\infty} \frac{\sin nx}{\ln(n+1)}$.

Pointwise Convergence of Fourier Series

Theorem 2. If f and f' are piecewise continuous on $[-L, L]$, then for any x in $(-L, L)$

$$(21) \quad \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\} = \frac{1}{2} [f(x^+) + f(x^-)],$$

where the a_n 's and b_n 's are given by the Euler-Fourier formulas (9) and (10). For $x = \pm L$, the series converges to $\frac{1}{2} [f(-L^+) + f(L^-)]$.[†]

In other words, when f and f' are piecewise continuous on $[-L, L]$, the Fourier series converges to $f(x)$ whenever f is continuous at x and converges to the average of the left- and right-hand limits at points where f is discontinuous.

Observe that the left-hand side of (21) is periodic of period $2L$. This means that if we extend $f(x)$ from the interval $(-L, L)$ to the entire real line using $2L$ -periodicity, then equation (21) holds for all x for the $2L$ -periodic extension of $f(x)$.

Example 6 To which function does the Fourier series for

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$

converge?

Solution In Example 4 we found that the Fourier series for $f(x)$ is given by (12), and in Figure 10.7, we sketched the graphs of two of its partial sums. Now $f(x)$ and $f'(x)$ are piecewise continuous in $[-\pi, \pi]$. Moreover, f is continuous except at $x = 0$. Thus, by Theorem 2, the Fourier series of f in (12) converges to the 2π -periodic function $g(x)$, where $g(x) = f(x) = -1$ for $-\pi < x < 0$, $g(x) = f(x) = 1$ for $0 < x < \pi$, $g(0) = [f(0^+) + f(0^-)]/2 = 0$, and at $\pm\pi$ we have $g(\pm\pi) = [f(-\pi^+) + f(\pi^-)]/2 = (-1 + 1)/2 = 0$. The graph of $g(x)$ is given in Figure 10.9. ♦

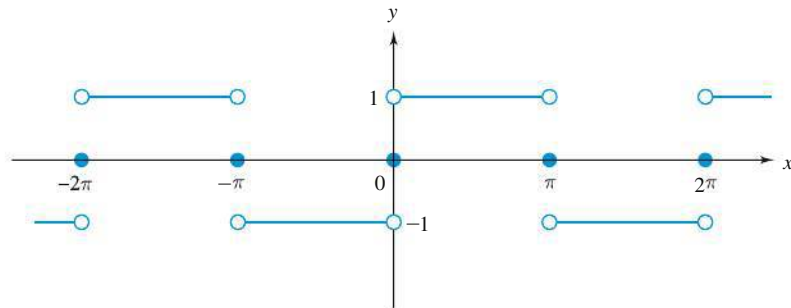


Figure 10.9 The limit function of the Fourier series for $f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi \end{cases}$

[†]From formula (21), we see that it doesn't matter how we define $f(x)$ at its points of discontinuity, since only the left- and right-hand limits are involved. The derivative $f'(x)$, of course, is undefined at such points.

When f is a $2L$ -periodic function that is continuous on $(-\infty, \infty)$ and has a piecewise continuous derivative, its Fourier series not only converges at each point—it converges **uniformly on** $(-\infty, \infty)$. This means that for any prescribed tolerance $\varepsilon > 0$, the graph of the partial sum

$$s_N(x) := \frac{a_0}{2} + \sum_{n=1}^N \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\}$$

will, for all N sufficiently large, lie in an ε -corridor about the graph of f on $(-\infty, \infty)$ (see Figure 10.10). The property of uniform convergence of Fourier series is particularly helpful when one needs to verify that a formal solution to a partial differential equation is an actual (genuine) solution.

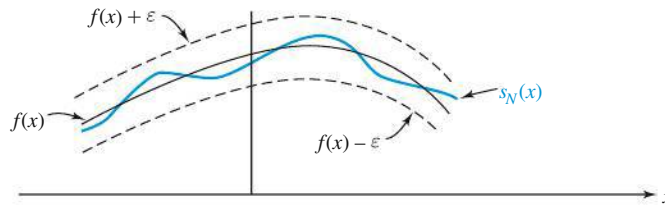


Figure 10.10 An ε -corridor about f

Uniform Convergence of Fourier Series

Theorem 3. Let f be a continuous function on $(-\infty, \infty)$ and periodic of period $2L$. If f' is piecewise continuous on $[-L, L]$, then the Fourier series for f converges uniformly to f on $[-L, L]$ and hence on any interval. That is, for each $\varepsilon > 0$, there exists an integer N_0 (that depends on ε) such that

$$\left| f(x) - \left[\frac{a_0}{2} + \sum_{n=1}^N \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\} \right] \right| < \varepsilon,$$

for all $N \geq N_0$, and all $x \in (-\infty, \infty)$.

In Example 5 we obtained the Fourier series expansion given in (13) for $f(x) = |x|$, $-1 < x < 1$. Since $g(x)$, the periodic extension of $f(x)$ (see Figure 10.11 on page 583) is continuous on $(-\infty, \infty)$ and

$$f'(x) = \begin{cases} -1, & -1 < x < 0, \\ 1, & 0 < x < 1, \end{cases}$$

is piecewise continuous on $[-1, 1]$, the Fourier series expansion (13) converges uniformly to $|x|$ on $[-1, 1]$. Compare Figure 10.8 on page 578.

The term-by-term differentiation of a Fourier series is not always permissible. For example, the Fourier series for $f(x) = x$, $-\pi < x < \pi$ (see Problem 9), is

$$(22) \quad f(x) \sim 2 \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n},$$

which converges for all x , whereas its derived series

$$2 \sum_{n=1}^{\infty} (-1)^{n+1} \cos nx$$

diverges for every x . The following theorem gives sufficient conditions for using termwise differentiation.

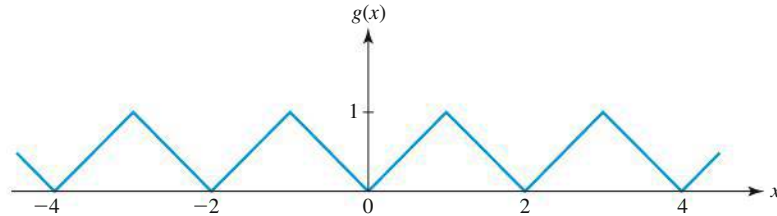


Figure 10.11 Periodic extension of $f(x) = |x|$, $-1 < x < 1$

Differentiation of Fourier Series

Theorem 4. Let $f(x)$ be continuous on $(-\infty, \infty)$ and $2L$ -periodic. Let $f'(x)$ and $f''(x)$ be piecewise continuous on $[-L, L]$. Then, the Fourier series of $f'(x)$ can be obtained from the Fourier series for $f(x)$ by termwise differentiation. In particular, if

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\},$$

then

$$f'(x) \sim \sum_{n=1}^{\infty} \frac{\pi n}{L} \left\{ -a_n \sin \frac{n\pi x}{L} + b_n \cos \frac{n\pi x}{L} \right\}.$$

Notice that Theorem 4 does not apply to the example function $f(x)$ and its Fourier series expansion shown in (22), since the 2π -periodic extension of this $f(x)$ fails to be continuous on $(-\infty, \infty)$.

Termwise integration of a Fourier series is permissible under much weaker conditions.

Integration of Fourier Series

Theorem 5. Let $f(x)$ be piecewise continuous on $[-L, L]$ with Fourier series

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right\}.$$

Then, for any x in $[-L, L]$,

$$\int_{-L}^x f(t) dt = \int_{-L}^x \frac{a_0}{2} dt + \sum_{n=1}^{\infty} \int_{-L}^x \left\{ a_n \cos \frac{n\pi t}{L} + b_n \sin \frac{n\pi t}{L} \right\} dt.$$

A final note on the convergence issue: The question as to under what conditions the Fourier series converges has led to a tremendous amount of beautiful mathematics. As a practical matter, however, time and economics will often dictate that you can afford to compute only a few terms of an eigenfunction series for your particular partial differential equation. Convergence fades into the background, and you need to know how to “do the best with what you’ve got.” So it is extremely gratifying to know that even if you’re going to use only part of an eigenfunction expansion, the Fourier coefficients (9)–(10) are still the best choice; Problem 37 demonstrates that every partial sum of a Fourier series outperforms any comparable superposition of the trigonometric functions, in terms of mean-square approximation.

10.3 EXERCISES

In Problems 1–6, determine whether the given function is even, odd, or neither.

1. $f(x) = x^3 + \sin 2x$
2. $f(x) = \sin^2 x$
3. $f(x) = (1 - x^2)^{-1/2}$
4. $f(x) = \sin(x + 1)$
5. $f(x) = e^{-x} \cos 3x$
6. $f(x) = x^{1/5} \cos x^2$

7. Prove the following properties:

- (a) If f and g are even functions, then so is the product fg .
- (b) If f and g are odd functions, then fg is an even function.
- (c) If f is an even function and g is an odd function, then fg is an odd function.

8. Verify formula (5). [Hint: Use the identity $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$.]

In Problems 9–16, compute the Fourier series for the given function f on the specified interval. Use a computer or graphing calculator to plot a few partial sums of the Fourier series.

9. $f(x) = x$, $-\pi < x < \pi$
10. $f(x) = |x|$, $-\pi < x < \pi$
11. $f(x) = \begin{cases} 1, & -2 < x < 0, \\ x, & 0 < x < 2 \end{cases}$
12. $f(x) = \begin{cases} 0, & -\pi < x < 0, \\ x^2, & 0 < x < \pi \end{cases}$
13. $f(x) = x^2$, $-1 < x < 1$
14. $f(x) = \begin{cases} x, & 0 < x < \pi, \\ x + \pi, & -\pi < x < 0 \end{cases}$
15. $f(x) = e^x$, $-\pi < x < \pi$
16. $f(x) = \begin{cases} 0, & -\pi < x < -\pi/2, \\ -1, & -\pi/2 < x < 0, \\ 1, & 0 < x < \pi/2, \\ 0, & \pi/2 < x < \pi \end{cases}$

In Problems 17–24, determine the function to which the Fourier series for $f(x)$, given in the indicated problem, converges.

17. Problem 9
18. Problem 10
19. Problem 11
20. Problem 12
21. Problem 13
22. Problem 14
23. Problem 15
24. Problem 16

25. Find the functions represented by the series obtained by the termwise integration of the given series from $-\pi$ to x .

$$(a) \quad 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx \sim x, \quad -\pi < x < \pi$$

$$(b) \quad \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{(2n+1)} \sim f(x),$$

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi \end{cases}$$

26. Show that the set of functions

$$\left\{ \cos \frac{\pi}{2}x, \sin \frac{\pi}{2}x, \cos \frac{3\pi}{2}x, \sin \frac{3\pi}{2}x, \dots, \cos \frac{(2n-1)\pi}{2}x, \sin \frac{(2n-1)\pi}{2}x, \dots \right\}$$

is an orthonormal system on $[-1, 1]$ with respect to the weight function $w(x) \equiv 1$.

27. Find the orthogonal expansion (generalized Fourier series) for

$$f(x) = \begin{cases} 0, & -1 < x < 0, \\ 1, & 0 < x < 1, \end{cases}$$

in terms of the orthonormal system of Problem 26.

28. (a) Show that the function $f(x) = x^2$ has the Fourier series, on $-\pi < x < \pi$,

$$f(x) \sim \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx.$$

- (b) Use the result of part (a) and Theorem 2 to show that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.$$

- (c) Use the result of part (a) and Theorem 2 to show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

29. In Section 8.8, it was shown that the Legendre polynomials $P_n(x)$ are orthogonal on the interval $[-1, 1]$ with respect to the weight function $w(x) \equiv 1$. Using the fact that the first three Legendre polynomials are

$$P_0(x) \equiv 1, \quad P_1(x) = x, \quad P_2(x) = (3/2)x^2 - (1/2),$$

find the first three coefficients in the expansion

$$f(x) = c_0 P_0(x) + c_1 P_1(x) + c_2 P_2(x) + \cdots,$$

where $f(x)$ is the function

$$f(x) := \begin{cases} -1, & -1 < x < 0, \\ 1, & 0 < x < 1. \end{cases}$$

30. As in Problem 29, find the first three coefficients in the expansion

$$f(x) = c_0 P_0(x) + c_1 P_1(x) + c_2 P_2(x) + \cdots,$$

when $f(x) = |x|$, $-1 < x < 1$.

31. The Hermite polynomials $H_n(x)$ are orthogonal on the interval $(-\infty, \infty)$ with respect to the weight function $W(x) = e^{-x^2}$. Verify this fact for the first three Hermite polynomials:

$$H_0(x) \equiv 1, \quad H_1(x) = 2x, \quad H_2(x) = 4x^2 - 2.$$

32. The Chebyshev (Tchebichef) polynomials $T_n(x)$ are orthogonal on the interval $[-1, 1]$ with respect to the weight function $w(x) = (1 - x^2)^{-1/2}$. Verify this fact for the first three Chebyshev polynomials:

$$T_0(x) \equiv 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1.$$

33. Let $\{f_n(x)\}$ be an orthogonal set of functions on the interval $[a, b]$ with respect to the weight function $w(x)$. Show that they satisfy the **Pythagorean property**

$$\|f_m + f_n\|^2 = \|f_m\|^2 + \|f_n\|^2$$

if $m \neq n$.

34. **Norm.** The norm of a function $\|f\|$ is like the length of a vector in R^n . In particular, show that the norm defined in (16) satisfies the following properties associated with length (assume f and g are continuous and $w(x) > 0$ on $[a, b]$):

- (a) $\|f\| \geq 0$, and $\|f\| = 0$ if and only if $f \equiv 0$.
 (b) $\|cf\| = |c| \|f\|$, where c is any real number.
 (c) $\|f + g\| \leq \|f\| + \|g\|$.

35. **Inner Product.** The integral in the orthogonality condition (14) is like the dot product of two vectors in R^n . In

particular, show that the **inner product** of two functions defined by

$$(23) \quad \langle f, g \rangle := \int_a^b f(x)g(x)w(x)dx,$$

where $w(x)$ is a positive weight function, satisfies the following properties associated with the dot product (assume f, g , and h are continuous on $[a, b]$):

- (a) $\langle f + g, h \rangle = \langle f, h \rangle + \langle g, h \rangle$.
 (b) $\langle cf, h \rangle = c \langle f, h \rangle$, where c is any real number.
 (c) $\langle f, g \rangle = \langle g, f \rangle$.

36. Complex Form of the Fourier Series.

- (a) Using the Euler formula $e^{i\theta} = \cos \theta + i \sin \theta$, $i = \sqrt{-1}$, prove that

$$\cos nx = \frac{e^{inx} + e^{-inx}}{2} \quad \text{and} \quad \sin nx = \frac{e^{inx} - e^{-inx}}{2i}.$$

- (b) Show that the Fourier series

$$\begin{aligned} f(x) &\sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\} \\ &= c_0 + \sum_{n=1}^{\infty} \{c_n e^{inx} + c_{-n} e^{-inx}\}, \end{aligned}$$

where

$$c_0 = \frac{a_0}{2}, \quad c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \frac{a_n + ib_n}{2}.$$

- (c) Finally, use the results of part (b) to show that

$$f(x) \sim \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad -\pi < x < \pi,$$

where

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

37. **Least-Squares Approximation Property.** The N th partial sum of the Fourier series

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos nx + b_n \sin nx\}$$

gives the best mean-square approximation of f by a trigonometric polynomial. To prove this, let $F_N(x)$ denote an arbitrary trigonometric polynomial of degree N :

$$F_N(x) = \frac{\alpha_0}{2} + \sum_{n=1}^N \{\alpha_n \cos nx + \beta_n \sin nx\},$$

and define

$$E := \int_{-\pi}^{\pi} [f(x) - F_N(x)]^2 dx,$$

which is the *total square error*. Expanding the integrand, we get

$$E = \int_{-\pi}^{\pi} f^2(x) dx - 2 \int_{-\pi}^{\pi} f(x) F_N(x) dx + \int_{-\pi}^{\pi} F_N^2(x) dx.$$

- (a) Use the orthogonality of the functions $\{1, \cos x, \sin x, \cos 2x, \dots\}$ to show that

$$\int_{-\pi}^{\pi} F_N^2(x) dx = \pi \left(\frac{\alpha_0^2}{2} + \alpha_1^2 + \dots + \alpha_N^2 + \beta_1^2 + \dots + \beta_N^2 \right)$$

and

$$\int_{-\pi}^{\pi} f(x) F_N(x) dx = \pi \left(\frac{\alpha_0 a_0}{2} + \alpha_1 a_1 + \dots + \alpha_N a_N + \beta_1 b_1 + \dots + \beta_N b_N \right).$$

- (b) Let E^* be the error when we approximate f by the N th partial sum of its Fourier series, that is, when we choose $\alpha_n = a_n$ and $\beta_n = b_n$. Show that

$$E^* = \int_{-\pi}^{\pi} f^2(x) dx - \pi \left(\frac{a_0^2}{2} + a_1^2 + \dots + a_N^2 + b_1^2 + \dots + b_N^2 \right).$$

- (c) Using the results of parts (a) and (b), show that $E - E^* \geq 0$, that is, $E \geq E^*$, by proving that

$$E - E^* = \pi \left\{ \frac{(\alpha_0 - a_0)^2}{2} + (\alpha_1 - a_1)^2 + \dots + (\alpha_N - a_N)^2 + (\beta_1 - b_1)^2 + \dots + (\beta_N - b_N)^2 \right\}.$$

Hence, the N th partial sum of the Fourier series gives the least total square error, since $E \geq E^*$.

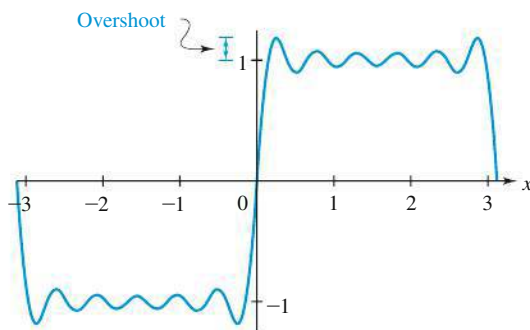
- 38. Bessel's Inequality.** Use the fact that E^* , defined in part (b) of Problem 37, is nonnegative to prove **Bessel's inequality**

$$(24) \quad \frac{a_0^2}{2} + \sum_{n=1}^{\infty} \{a_n^2 + b_n^2\} \leq \frac{1}{\pi} \int_{-\pi}^{\pi} f^2(x) dx.$$

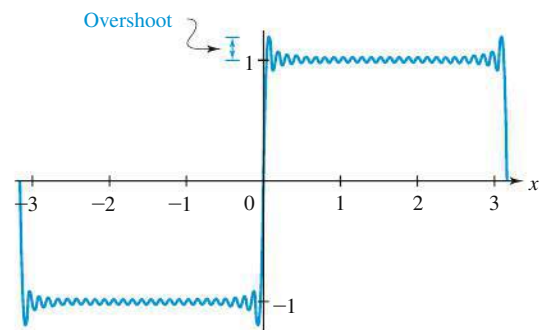
(If f is piecewise continuous on $[-\pi, \pi]$, then we have equality in (24). This result is called **Parseval's identity**.)

- 39. Gibbs Phenomenon.**[†] Josiah Willard Gibbs, who was awarded the first American doctorate in engineering (Yale, 1863), observed that near points of discontinuity of f , the partial sums of the Fourier series for f may overshoot by approximately 9% of the jump, regardless of the number of terms. This is illustrated in Figure 10.12 for the function

$$f(x) = \begin{cases} -1, & -\pi < x < 0, \\ 1, & 0 < x < \pi, \end{cases}$$



(a) Graph of $f_{11}(x)$



(b) Graph of $f_{51}(x)$

Figure 10.12 Gibbs phenomenon for partial sums of Fourier series

[†]*Historical Footnote:* Actually, H. Wilbraham discovered this phenomenon some 50 years earlier than Gibbs did. It is more appropriately called the *Gibbs–Wilbraham* phenomenon.

whose Fourier series has the partial sums

$$f_{2n-1}(x) = \frac{4}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \cdots + \frac{\sin(2n-1)x}{(2n-1)} \right].$$

To verify this for $f(x)$, proceed as follows:

(a) Show that

$$\begin{aligned} \pi(\sin x)f'_{2n-1}(x) &= 4 \sin x [\cos x + \cos 3x \\ &\quad + \cdots + \cos(2n-1)x] \\ &= 2 \sin 2nx. \end{aligned}$$

(b) Infer from part (a) and the figure that the maximum occurs at $x = \pi/(2n)$ and has the value

$$f_{2n-1}\left(\frac{\pi}{2n}\right) = \frac{4}{\pi} \left[\sin \frac{\pi}{2n} + \frac{1}{3} \sin \frac{3\pi}{2n} + \cdots + \frac{1}{2n-1} \sin \frac{(2n-1)\pi}{2n} \right].$$

(c) Show that if one approximates

$$\int_0^\pi \frac{\sin x}{x} dx$$

using the partition $x_k := (2k-1)(\pi/2n)$, $k = 1, 2, \dots, n$, $\Delta x_k = \pi/n$ and choosing the midpoint of

each interval as the place to evaluate the integrand, then

$$\begin{aligned} \int_0^\pi \frac{\sin x}{x} dx &\approx \frac{\sin(\pi/2n)}{\pi/2n} \frac{\pi}{n} + \cdots \\ &\quad + \frac{\sin[(2n-1)\pi/2n]}{(2n-1)\pi/2n} \frac{\pi}{n} \\ &= \frac{\pi}{2} f_{2n-1}\left(\frac{\pi}{2n}\right). \end{aligned}$$

(d) Use the result of part (c) to show that the overshoot satisfies

$$\lim_{n \rightarrow \infty} f_{2n-1}\left(\frac{\pi}{2n}\right) = \frac{2}{\pi} \int_0^\pi \frac{\sin x}{x} dx.$$

(e) Using the result of part (d) and a numerical integration algorithm (such as Simpson's rule, Appendix C) for the **sine integral function**

$$Si(z) := \int_0^z \frac{\sin x}{x} dx,$$

show that $\lim_{n \rightarrow \infty} f_{2n-1}(\pi/(2n)) \approx 1.18$. Thus, the approximations overshoot the true value of $f(0^+) = 1$ by 0.18, or 9% of the jump from $f(0^-)$ to $f(0^+)$.

10.4 Fourier Cosine and Sine Series

A typical problem encountered in using separation of variables to solve a partial differential equation is the problem of representing a function defined on some finite interval by a trigonometric series consisting of only sine functions or only cosine functions. For example, in Section 10.2, equation (25), page 569, we needed to express the initial values $u(x, 0) = f(x)$, $0 < x < L$, of the solution to the initial-boundary value problem associated with the problem of a vibrating string as a trigonometric series of the form

$$(1) \quad f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right).$$

Recalling that the Fourier series for an odd function defined on $[-L, L]$ consists entirely of sine terms, we might try to achieve (1) by artificially extending the function $f(x)$, $0 < x < L$, to the interval $(-L, L)$ in such a way that the extended function is odd. This is accomplished by defining the function

$$f_o(x) := \begin{cases} f(x), & 0 < x < L, \\ -f(-x), & -L < x < 0, \end{cases}$$

and extending $f_o(x)$ to all x using $2L$ -periodicity.[†] Since $f_o(x)$ is an odd function, it has a Fourier series consisting entirely of sine terms. Moreover, $f_o(x)$ is an extension of $f(x)$,

[†]Strictly speaking, we have extended $f_o(x)$ to all x other than the integer multiples of L . Continuity considerations often suggest appropriate values for the extended functions at some of these points, as well. Figure 10.13 on page 589 illustrates this.